



### Chapter 3: *Determinants*

#### Question 1:

Compute the determinants by cofactor expansions. At each step, choose a row or column that involves the least amount of computation.

a)

$$\begin{vmatrix} 6 & 0 & 0 & 5 \\ 1 & 7 & 2 & -5 \\ 2 & 0 & 0 & 0 \\ 8 & 3 & 1 & 8 \end{vmatrix}$$

b)

$$\begin{vmatrix} 1 & -2 & 5 & 2 \\ 0 & 0 & 3 & 0 \\ 2 & -6 & -7 & 5 \\ 5 & 0 & 4 & 4 \end{vmatrix}$$

#### Question 2:

Compute the determinant of the following matrix:

$$\begin{vmatrix} 4 & 0 & 0 & 0 \\ 7 & -1 & 0 & 0 \\ 2 & 6 & 3 & 0 \\ 5 & -8 & 4 & -3 \end{vmatrix}$$

#### Question 3:

Combine the methods of row reduction and cofactor expansion to compute the determinants.

a)

$$\begin{vmatrix} 2 & 5 & -3 & -1 \\ 3 & 0 & 1 & -3 \\ -6 & 0 & -4 & 9 \\ 4 & 10 & -4 & -1 \end{vmatrix}$$

b)

$$\begin{vmatrix} -1 & 2 & 3 & 0 \\ 3 & 4 & 3 & 0 \\ 5 & 4 & 6 & 6 \\ 4 & 2 & 4 & 3 \end{vmatrix}$$

Question 4:

Use the determinant to check whether the set of vectors is linearly independent:

a)  $v_1 = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}, v_2 = \begin{bmatrix} 3 \\ 3 \\ 2 \end{bmatrix}$  and  $v_3 = \begin{bmatrix} 0 \\ 4 \\ 1 \end{bmatrix}$

b)  $v_1 = \begin{bmatrix} 5 \\ 1 \\ 0 \end{bmatrix}, v_2 = \begin{bmatrix} 0 \\ -3 \\ 5 \end{bmatrix}$  and  $v_3 = \begin{bmatrix} -1 \\ -2 \\ 3 \end{bmatrix}$

Question 5:

Find the determinants where:

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = 7.$$

a)

$$\begin{vmatrix} a & b & c \\ d & e & f \\ 3g & 3h & 3i \end{vmatrix}$$

b)

$$\begin{vmatrix} a & b & c \\ 5d & 5e & 5f \\ g & h & i \end{vmatrix}$$

c)

det of

$$\begin{bmatrix} a+d & b+e & c+f \\ d & e & f \\ g & h & i \end{bmatrix}$$

d)

$$\begin{vmatrix} d & e & f \\ a & b & c \\ g & h & i \end{vmatrix}$$

Question 6:

Let A and B be 3 x 3 matrices, with  $\det A = 4$  and  $\det B = -3$ . Use properties of determinants to compute.

a)  $\det AB$

b)  $\det 5A$

c)  $\det B^T$

d)  $\det A^T B A$

e)  $\det B^{-1} A B$

Question 7:

Use Cramer's rule to compute the solutions of the systems in:

$$\text{a) } \begin{cases} 5x_1 + 7x_2 = 3 \\ 2x_1 + 4x_2 = 1 \end{cases}$$

$$\text{b) } \begin{cases} 4x_1 + x_2 = 6 \\ 5x_1 + 2x_2 = 7 \end{cases}$$

$$\text{c) } \begin{cases} 3x_1 - 2x_2 = 7 \\ -5x_1 + 6x_2 = -5 \end{cases}$$

$$\text{d) } \begin{cases} -5x_1 + 3x_2 = 9 \\ 3x_1 - x_2 = -5 \end{cases}$$

Question 8:

Compute the adjugate of the given matrix and then use the theorem 8 to give the inverse of the matrix:

$$\text{a) } \begin{bmatrix} 0 & -2 & -1 \\ 3 & 0 & 0 \\ -1 & 1 & 1 \end{bmatrix}$$

$$\text{b) } \begin{bmatrix} 1 & 1 & 3 \\ 2 & -2 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

Question 9:

Find the volume of the parallelepiped with one vertex at the origin and adjacent vertices at  $(1,0,-2)$ ,  $(1,2,4)$ , and  $(7,1,0)$ .

Question 10:

Let  $S$  be the parallelogram determined by the vectors  $b_1 = \begin{bmatrix} -3 \\ 2 \end{bmatrix}$  and  $b_2 = \begin{bmatrix} -3 \\ 6 \end{bmatrix}$  and  $b_3 = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$  and let  $A = \begin{bmatrix} 6 & -2 \\ -3 & 2 \end{bmatrix}$ . Compute the area of the image of  $S$  under the mapping  $x \rightarrow Ax$ .